



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2025– 2026 (Odd Semester)

Degree, Semester & Branch : V Semester B.E CSE
Course Code & Title : CCS366 & Software Testing and Automation
Name of the Faculty member : Mrs. S. Vijaya Amala Devi, AP / CSE

Innovative Practice Description

- **Unit / Topic:** Unit I/FOUNDATION OF SOFTWARE TESTING/ Program Correctness and Verification, Reliability versus Safety
- **Course Outcome:** CO1
- **Topic Learning Outcome:** 1a
- **Activity Chosen:** Think-Pair-Share
- **Justification**
- **Program Correctness and Verification** and **Reliability versus Safety** are topics that need careful thinking and clear understanding because they involve logic, correctness proofs, and comparing system behaviors.
- Using **Think–Pair–Share** helps students first think on their own, then discuss their ideas with a partner, which makes it easier for them to understand the concepts and correct any misunderstandings.
- When students share their ideas with the class, they hear different viewpoints, learn clearer explanations, and strengthen their understanding. This method increases participation, improves clarity, and makes difficult concepts easier to understand through discussion instead of only listening to lectures.
- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**
- Think–Pair–Share innovative practice was conducted for III year CSE students after the explanation of the concepts of *Program Correctness and Verification* and *Reliability versus Safety*. Students were introduced to how correctness ensures that a program meets its specifications and how reliability and safety differ in terms of system performance under failures.
- I initiated the activity by asking students to reflect individually on the meaning of program correctness, encouraging them to think about preconditions, postconditions, invariants, and the need for verifying whether a program behaves as expected.
- Similarly, they were asked to consider the difference between reliability (consistent performance over time) and safety (protection from harmful consequences of failures). Students were prompted to think about simple example such as online transaction systems, and safety-critical system to connect the concepts with real-life scenarios.
- After students completed their individual thinking, they were grouped into pairs and asked to share and compare their ideas. Each pair discussed how correctness can be proven with logical reasoning and why reliability and safety must be evaluated differently in real

systems. They exchanged their understanding of failure handling, error detection, and the importance of system guarantees.

- Finally, Mr. Janaramji pairs of were asked to explain their ideas to the whole class for broader discussion. Students presented how correctness verification helps prevent logical errors and how reliability ensures consistent behavior, while safety prevents harmful outcomes.
- They shared simple examples to demonstrate the differences between the concepts. This classroom sharing enriched everyone's understanding and reinforced the key ideas through peer explanation and collaborative learning.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PO12	PSO1
CO5	3	3	3	1	1	1	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovation Practice	PO1	PO2	PO3	PO4	PO12	PSO1
	2	2	3	2	2	2
Justification for Correlation	Students able to understand software testing concepts requires a solid foundation in engineering principles to grasp the technical aspects of testing methodologies and tools.	Students able to analyze the need for software testing involves identifying potential issues and risks in software systems, which is essential for effective problem-solving.	Students able to recognize the need for software testing involves considering testing strategies and techniques during the design and development phases to ensure the delivery of reliable solutions.	Students able to investigate complex issues and identifying their root causes, which requires analytical skills and problem-solving abilities.	Students able to recognize the dynamic nature of software testing, engineers should cultivate a mindset of continuous learning to stay updated with evolving testing methodologies and technologies.	Students able to Interpret software testing concepts aligns with proficiency in software development, ensuring thorough testing of complex mobile applications.

- **Images / Screenshot of the practice:**



Figure1: Students discussing and presenting their ideas with team members.



Figure2: Students engaged in discussion with their teams.

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Collaborating in pairs enables individuals to exchange their expertise, leading to a more comprehensive grasp of the topic through shared insights. This collaborative approach enhances understanding by leveraging the collective knowledge of both individuals.

❖ **Feedback of practice from students and other stakeholders:**

- This activity served as an opportunity for students to review and reinforce their understanding of Program Correctness and Verification as well as Reliability versus Safety. It allowed them to revisit key ideas such as preconditions, postconditions, invariants, system reliability, and safety constraints, helping them focus on areas that needed clarification.
- The activity helped the instructor analyze, evaluate, and enhance students' learning progress. By observing student discussions and explanations, the instructor gained insight into how well students understood correctness proofs, verification logic, and the distinction between reliability and safety.
- Various types of conceptual and scenario-based questions were discussed in the classroom, giving students a comprehensive understanding of how to evaluate correctness, verify program behavior, and differentiate system reliability from safety in real-world applications.

❖ **Benefit of the practice:**

- The students are actively involved in the learning process, and there is a mutual exchange of knowledge among them. Through discussion of program correctness, verification conditions, reliability measures, and safety requirements, learners gain a clearer understanding by sharing different viewpoints.
- Promotes an interactive classroom environment that cultivates a friendly yet academically-focused atmosphere. Students feel comfortable discussing logical reasoning, correctness proofs, and real-time system behavior in pairs before speaking to the class.
- Incorporates all students into the teaching–learning process. Even hesitant learners are encouraged to participate through guided thinking, peer discussion, and collaborative explanation of correctness and safety concepts.

❖ **Challenges faced in implementation**

- Students encountered significant difficulty with this activity and struggled to complete it within the given time frame, particularly when dealing with new real-time system concepts. Topics such as preconditions, invariants, proof techniques, and differences between reliability and safety required additional time and support for full understanding.
- Students exhibited hesitancy when it came to speaking in front of the entire class. Sharing correctness proofs or reliability–safety interpretations made some students nervous, affecting their confidence during whole-class discussions.

References:

- ❖ <http://www.et.iitb.ac.in/TeachingResources.html>
- ❖ <https://www.readingrockets.org/strategies/think-pair-share>
- ❖ <https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improve-your-classroom-discussions1704.html>
- ❖ <https://serc.carleton.edu/introgeo/interactive/tpshare.html>

Signature of Faculty Member

HOD